

ABSTRACT

The Human Resource (HR) Office of Ilocos Sur Polytechnic State College (ISPSC) Tagudin Campus maintained its employee records on paper and in Excel workbooks, keeping a Personal Data Sheet for every employee, a separate form for every leave application, and two handwritten ledger cards per person, while leave approval required physically routing a printed form between three offices. This study developed NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College, a web-based system that moves this workflow online while preserving the Civil Service forms exactly as the campus uses them. The system was developed using the Laravel framework, MySQL, and Blade with Tailwind CSS, and follows the Rapid Application Development (RAD) model. NexHRIS provides employee records management, Personal Data Sheet submission and review, a role-derived leave approval chain, leave and service credit ledger cards, HR policies with acknowledgement tracking, announcements, a digital identification card, role-based dashboards, and an audit trail. A distinguishing feature of the system is that official documents are produced by converting the actual uploaded workbook rather than rebuilding the form from database values, preserving the exact format of the campus's Civil Service forms. Access is governed by four roles — HR Administrator, Campus Director, Dean, and Employee — with data boundaries enforced at the query level. The acceptability of the system was evaluated using the Technology Acceptance Model (TAM) in terms of Perceived Usefulness and Perceived Ease of Use, and the Wilcoxon Signed-Rank Test was used to determine whether a significant difference existed between the existing manual process and the developed system.

Keywords: Human Resource Information System, Personal Data Sheet, Leave Management, Role-Based Access Control, Technology Acceptance Model, ISPSC

Chapter 1

INTRODUCTION

Background of the Study

The rapid advancement of technology has fundamentally transformed the way organizations manage their human resources. Across the globe, institutions are shifting away from manual, paper-based administrative processes toward integrated digital platforms that offer efficiency, accuracy, and accessibility. In government higher education institutions, Human Resource (HR) offices are primarily responsible for maintaining, organizing, and safeguarding the employment records of all personnel. These records, collectively known as the 201 files, contain an employee's complete employment history, including personal information, appointment papers, service records, certificates of eligibility, and other official employment documents. The proper management and accessibility of these records are critical to the effective administration of human resources within any institution.

Internationally, several studies confirm the urgent need for digital employee document management systems in organizations. Wahdaniah et al. (2023) demonstrated that human resource management transformation in the digital age results in more organized recordkeeping, reduced document retrieval time, and improved overall HR operational efficiency across diverse institutional settings. Similarly, Abaci and Medeni (2022) conducted a case study on the efficiency of electronic document management systems and established that digitizing employee records reduces retrieval time, minimizes physical storage requirements, and enhances organizational transparency. Orioque et al. (2024) further affirmed that an online contextualized document management system aligned with an organization's operational environment

significantly enhances information access and reduces misplaced documentation. Dres and Rabut (2025) developed eDOCS, a web-based file management system, and found it to be highly effective in improving document organization, accessibility, and retrieval speed compared to traditional manual filing methods. Furthermore, Panfilova et al. (2024) noted that the integration of automated document management systems into institutional workflows facilitates diagnostic monitoring, reduces human error, and improves documentation lifecycle management, all of which are directly relevant to the employee file management needs of human resource offices in academic institutions.

In the Philippines, the digitalization of employee and institutional records has similarly gained traction among higher education institutions. Purcia and Velarde (2022) conducted a study on student registration and records management in private universities and found that the majority of institutions still rely on paper-based processes, highlighting a persistent gap in digital adoption and the urgent need for electronic records systems. Pagayonan (2022) developed an Electronic Document Management System (eDMS) for a higher education institution and found that the system improved the organization, storage, and retrieval of institutional records while reducing dependency on physical filing systems. Encarnacion et al. (2024) developed a web-based document management system for an educational office in Pagadian City that streamlined the dissemination, submission, and collection of pertinent documents, addressing challenges typically experienced by office employees in managing paper-based records. More recently, Rabulan and Glina (2025) developed a web-based file management system for the Office of National Service Training Program at Sorsogon State University, which effectively centralized student information and office records, resulting in a more secure, accurate, and faster file management process that directly reduced the burden on office personnel.

At the regional and provincial level, the need for digital records management has been consistently identified, particularly at ISPSC Tagudin Campus itself. Angala et al. (2023) developed and implemented a web-based Document Management System (DMS) at ISPSC Tagudin Campus and found that respondents were generally very satisfied with how the system was used to arrange, store, track, and manage documents within the campus, affirming that digital document systems are well-received at the institution. Bulong et al. (2025) similarly focused on ISPSC Tagudin Campus and developed DocuTrack, a web-based document management system featuring role-based access control, cloud storage integration, and an activity log system, confirming that paper-based document management remains a critical challenge for the institution as it continues to grow in terms of personnel, programs, and administrative responsibilities. Beyond ISPSC, Sebucao-Orias (2023) examined records management practices of a state university in the Philippines and identified significant deficiencies in documentation processes, while Taguiam (2024) established that non-compliance with proper records management protocols in state universities leads to operational inefficiencies, both underscoring the need for structured digital records management systems in similar institutions. The Ilocos Regional Development Plan 2023-2028 further identifies the enhancement of institutional efficiency and the adoption of digital governance tools as priority directions for all government agencies and institutions in the region (Regional Development Council, 2023). The Data Privacy Act of 2012 (Republic Act 10173) likewise mandates that all government institutions, including state colleges in Ilocos Sur, ensure the confidentiality, integrity, and availability of personal data in their custody, which directly necessitates the adoption of secure digital systems for managing sensitive employee records. Furthermore, the national directive on digital transformation articulated in the Philippine Development Plan 2023-2028

reinforces the importance of transitioning from paper-based to digital HR record management systems in institutions such as ISPSC (Department of Information and Communications Technology, 2023).

The implementation of NexHRIS at Ilocos Sur Polytechnic State College, Tagudin Campus is particularly justified by the institution's current state of HR document management. The ISPSC HR Office presently maintains all employee 201 files on paper and in Excel workbooks, keeping a Personal Data Sheet per employee, a separate form per leave application, and two handwritten ledger cards per person. Leave approval requires physically routing a printed form between three offices for signature. These manual systems are prone to document loss, physical deterioration, unauthorized access, and delays in document retrieval. As the institution continues to grow in terms of personnel and administrative responsibilities, the volume of employee records that must be managed likewise increases, making traditional paper-based processes increasingly inefficient and difficult to sustain. The absence of a centralized digital repository for employee files requires HR personnel to spend considerable time locating, reproducing, and verifying documents that could otherwise be accessed instantly through a secure electronic system. Employees, on their end, must still physically visit the HR Office just to update their Personal Data Sheet or to check their leave credits and file a leave application, adding to the workload of HR personnel who have to manually retrieve records and compute balances for every request.

A review of the systems cited above reveals a consistent limitation: all of them are document management systems, not human resource systems. The eDMS of Pagayonan (2022), eDOCS of Dres and Rabut (2025), the contextualized system of Orioque et al. (2024), the web-based system of Encarnacion et al. (2024), and the file management system of Rabulan and Glina (2025) are each designed to store, categorize,

track, and retrieve files. The same is true of the two systems developed for ISPSC Tagudin Campus itself: the DMS of Angala et al. (2023) and DocuTrack by Bulong et al. (2025) both centralize institutional documents, and DocuTrack adds role-based access and activity logging, but both stop at document handling. In every one of these systems, a document is treated as a file to be uploaded and found again, and none of them process the transaction that the document represents.

NexHRIS differs from all of them in four specific ways. First, none of the reviewed systems handle the Personal Data Sheet as a governed submission with a review-and-return cycle and retained versions; they would treat a PDS as one more uploaded file. Second, none of them route a document through a multi-level approval chain in which the route is determined by the rank of the applicant, nor do they record which approvals were skipped and why. Third, none of them maintain a running-balance record such as the campus's leave and service credit ledger cards, where correcting one entry must replay every balance below it. Fourth, none of them are concerned with preserving the exact printed format of an official government form, because a document management system stores the file the user uploaded and never has to regenerate it. These are precisely the day-to-day HR transactions that ISPSC employees still complete manually, even though the campus already has a working document management system. The gap, therefore, is not the digitization of storage, which the literature has addressed thoroughly, but the digitization of the HR transactions themselves — which is what NexHRIS is designed to fill.

Recognizing this gap, NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College is conceptualized to go beyond the storage-and-retrieval model demonstrated by existing systems. Rather than covering the entire employee 201 file, the study focuses

specifically on the components employees interact with most directly and repeatedly — the Personal Data Sheet, leave records, personal information, and HR policies and guidelines — allowing employees to take part in managing their own HR transactions and staying informed of institutional policies, instead of relying solely on manual, HR-mediated processes. Through the development of NexHRIS, the study seeks to address the limitations of existing manual practices, strengthen records security and accessibility, improve operational efficiency, and support compliance with government policies on digital transformation and data privacy. Ultimately, the study aims to contribute to the modernization of human resource management practices in ISPSC and provide a model for the implementation of HR-focused information systems in higher education institutions.

Conceptual Framework of the Study

The conceptual framework of this study is anchored on the Input–Process–Output (IPO) model, which provides a systematic representation of how data is transformed into meaningful outputs through the proposed system.

The Input consists of the employee 201 files, specifically the Personal Data Sheet (PDS) and leave records, together with employees' personal information and the HR policies and guidelines maintained by the Human Resource (HR) Office. These constitute the primary data required for managing employee records and facilitating HR-related transactions within the system.

The Process involves the design, development, and implementation of NexHRIS. During this stage, the system validates and updates employee information, routes leave applications through a role-derived approval chain, records approved leave against the appropriate ledger card, organizes employee data, and manages HR policies

and guidelines through a centralized digital platform. These processes transform the input data into accurate, organized, and readily accessible information.

The Output consists of a centralized digital employee records repository, online Personal Data Sheet management, digital leave records and leave application management, maintained leave and service credit ledger cards, a digital employee identification record, and accessible HR policies and guidelines. Collectively, these outputs enhance the security, accessibility, and efficiency of employee records management while improving the delivery of HR services.

Overall, the IPO model illustrates how NexHRIS transforms manually managed employee records and HR information into an integrated digital system that supports efficient record management, facilitates HR transactions, and promotes effective human resource administration at Ilocos Sur Polytechnic State College.

[Figure 1. Research Paradigm (Input – Process – Output diagram) — insert screenshot/image here]

Figure 1. Research Paradigm

Objectives of the Study

The main objective of this study is to develop NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College to support the secure and efficient digital management of employee Personal Data Sheets (PDS) and leave records at ISPSC Tagudin Campus. Specifically, this study aims:

1. To analyze the current employee file management processes of the ISPSC HR Office to serve as the basis for the system design and development;
2. To determine the system requirements for the development of NexHRIS, a web-based Human Resource Information System for the secure

management of employee Personal Data Sheets, leave records, and personal information at ISPSC Tagudin Campus;

3. To evaluate the level of acceptability of NexHRIS using the Technology Acceptance Model (TAM); and
4. To determine whether there is a significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use.

Hypothesis

H₀: There is no significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use.

Scope and Limitation of the Study

This study covers the design, development, and implementation of NexHRIS for ISPSC Tagudin Campus, focusing on four components of the employee 201 file: the Personal Data Sheet, leave records, personal information, and HR policies. The HR Administrator publishes the official Civil Service forms; employees download, fill, and upload them, and the HR Administrator reviews each submission and either approves it or returns it with remarks, retaining every version. Leave applications are routed through an approval chain determined by the rank of the applicant, since no user signs their own leave: an employee's application goes from the Dean, to the HR Administrator, to the Campus Director; a Dean's skips the Dean stage; and the Campus Director's goes to the HR Administrator alone, as no role outranks the Campus Director. Skipped stages still appear on the printed trail, marked not applicable. Once an application is fully approved, the HR Administrator records the days against either the leave ledger or the service credit ledger. The system serves four roles — HR

Administrator, Campus Director, Dean, and Employee — whose data boundaries are enforced at the query level rather than in the interface, so a user cannot reach records outside their authority by modifying a web address. It also provides each employee a digital identification card with a QR code, published HR policies with acknowledgement tracking, announcements, role-based dashboards, and an audit trail. NexHRIS is web-based and accessible from any internet-connected device.

The study is limited in the following respects. It does not digitize the rest of the 201 file — appointment papers, service records, performance ratings, and similar documents remain in the manual custody of the HR Office. The system does not perform optical character recognition or extract data from uploaded documents, and does not verify the accuracy or authenticity of what an employee encodes, which remains the responsibility of the HR Administrator on review. Leave credits and ledger balances are encoded and maintained by the HR Administrator; the system neither computes credits from attendance or biometric data nor decides which of the two ledger cards an approved leave is recorded on. The policies module supports viewing and acknowledgement only, without a facility for employees to comment on or request revisions to a policy. The digital identification card is viewable and printable within the system and carries no RFID functionality. NexHRIS does not integrate with external government systems such as the Civil Service Commission, Government Service Insurance System, or Department of Budget and Management; is intended for ISPSC Tagudin Campus alone and does not support multi-campus deployment; has no dedicated mobile application, though it may be opened in a mobile browser; and requires an internet connection, with no offline operation.

Importance of the Study

The development of NexHRIS is expected to provide significant benefits to various stakeholders by improving the management, accessibility, and security of employee records. The beneficiaries of this study are discussed as follows:

To the HR Office of ISPSC Tagudin Campus. The NexHRIS system will provide a centralized and secure platform for managing employee 201 files and personnel records. It will reduce the time and effort required to store, retrieve, and verify employee documents, and it automates the routing of leave applications and the maintenance of ledger cards, thereby improving operational efficiency and supporting more effective human resource management.

To the Employees of ISPSC Tagudin Campus. The system will provide employees with a transparent and accessible platform where they can independently submit and print their own Personal Data Sheet, file and monitor their leave applications, view their own ledger cards, and view their own personal information, reducing the need for repeated visits to the HR Office for these routine transactions.

To Ilocos Sur Polytechnic State College (ISPSC). The institution will benefit from improved records security, reduced risk of document loss or damage, and a more organized employee records management process. The system will also support compliance with the Data Privacy Act of 2012 and the institution's digital transformation initiatives.

To the Researchers. The study will provide practical experience in designing, developing, and implementing a web-based information system that addresses an actual institutional need. It will also enhance the researchers' knowledge and skills in system development, document processing, and system evaluation.

To Future Researchers. This study may serve as a valuable reference for future researchers who intend to develop similar human resource information systems or

employee records management solutions. The findings and methodologies presented may contribute to further research and innovation in the field of HR technology.

Chapter 2

METHODOLOGY

This chapter presents the methodology to be employed in the development of NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College. It discusses the research design, software development model, project timeline, project assignments, target population and locale, research instruments, and the methods of data analysis to be used in the study.

Research Design

This study will use a descriptive developmental research design. The descriptive component will involve gathering data on the current employee file management practices and challenges of the ISPSC HR Office through interviews and observations, which will serve as the basis for identifying the required system features and design specifications. The developmental component will focus on the actual design, development, and implementation of NexHRIS as a functional web-based employee file management system tailored to the needs of the HR Office and employees of ISPSC Tagudin Campus.

According to Kosie and Lew-Williams (2022), developmental research is the systematic process of designing, developing, and evaluating programs or systems to ensure internal consistency and effectiveness. The descriptive component documents each stage of development, while the developmental component emphasizes iterative enhancement and continuous refinement based on user needs and system testing results. This design framework will guide the entire research process from the initial requirements-gathering phase through system development, testing, and final evaluation.

In the context of this study, the descriptive developmental design will begin with an analysis of the current HR document management process at ISPSC Tagudin Campus, followed by the identification of the specific features required in NexHRIS based on the actual needs of the HR Administrator and employees. The system will then be designed, developed, and iteratively tested, with refinements made based on user feedback, until the final version is deployed and formally evaluated.

Software Model

This study will adopt the Rapid Application Development (RAD) Model as the software development methodology. RAD is an incremental development model that prioritizes rapid prototyping, iterative delivery, and continuous user feedback over lengthy planning and documentation phases. It is particularly well-suited for developing institutional systems where user requirements are well-defined and where the scope allows for modular development and frequent user involvement in the refinement process (Pressman, 2020).

In the context of this study, the RAD model will support the fast-paced development of NexHRIS by allowing each component — employee PDS management, the leave application and approval chain, ledger card maintenance, and personal information management — to be built and tested in stages, with feedback from the HR Administrator incorporated at each phase before proceeding to the next.

[Figure 2. Rapid Application Development — insert screenshot/image here]

Figure 2. Rapid Application Development

Requirements Planning. The researchers will conduct structured interviews and consultations with the ISPSC HR Administrator to identify the current challenges in employee file management and define the functional requirements of NexHRIS. These requirements will include the need for a centralized repository for PDS and leave

records, a secure role-based access system, a multi-level leave filing and monitoring feature, and the faithful reproduction of the campus's official Civil Service forms.

User Design. A user-centered approach will be applied to design wireframes and interface layouts for all system modules. The HR Administrator's feedback will guide the design of the PDS review panel, the leave application review interface, the ledger card maintenance interface, and the employee account management panel, while employee user experience will be considered in the design of the PDS submission dashboard, the leave filing and monitoring interface, and the personal information view.

Construction. The system will be built using the Laravel PHP framework for the backend, Blade with Tailwind CSS and Alpine.js for the frontend interface, and MySQL as the relational database management system. PhpSpreadsheet is used to read the uploaded Civil Service workbooks, while document conversion to PDF is handled by LibreOffice where it is available and by a pure-PHP path using PhpSpreadsheet and Dompdf where it is not. Chart.js is used for dashboard visualizations. Modules will be developed sequentially, with each module tested and refined before integration with the rest of the system.

Cutover. The completed NexHRIS system will be deployed on a shared hosting environment. During development the system was hosted on Hostinger's shared hosting with cloud-based document storage, while the final deployment target is a cPanel shared hosting environment running PHP 8.2 and MySQL; the system is designed to run in either environment because it automatically selects a document renderer based on what the host supports. This phase will include system testing, user orientation and training for the HR Administrator and selected employee respondents, final adjustments based on evaluation feedback, and the formal administration of the Technology Acceptance Model (TAM) questionnaire.

Project Plan

The development of NexHRIS will follow the four phases of the Rapid Application Development (RAD) Model. The project is planned across a period covering June 2026 to September 2026, with each phase assigned specific milestones and deliverables to ensure timely and organized system development.

[Figure 3. Project Schedule (Gantt chart) — insert screenshot/image here]

Figure 3. Project Schedule

Project Assignment

The roles and responsibilities of the project team for NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College are presented in Table 1. Each member is assigned specific tasks based on their skills and areas of specialization to ensure proper development and completion of the system.

Table 1. Project Assignment

Roles	Name	Function
Project Manager	Eidref Jake S. Manalansan	Responsible for coordinating with the project team; maintains open communication with all the members.
System Analyst and Designer	Eidref Jake S. Manalansan; Diana Sofia L. Sta. Ana	Coordinate the technical team's efforts in resolving challenges and ensuring that solutions are practical and consistent.
Programmer and Developer	Eidref Jake S. Manalansan	Manage the systems planning and design team and build the project.
QA / Tester	Diana Sofia L. Sta. Ana; Sheran S. Leal	Responsible for checking the debugging queries of the project; evaluate software needs and communicate findings to the project team.
Documenter / Technical Writer	Eidref Jake S. Manalansan; Diana Sofia L. Sta. Ana; Sheran S. Leal; Yuri Bib P. Somera; Arvie Keith M. Manzano	Publish documents to the respective parties; overall project status reporting.

Population and Locale of the Study

The study will be conducted at Ilocos Sur Polytechnic State College (ISPSC), Tagudin Campus, located in Tagudin, Ilocos Sur. This institution serves as the locale of the study, as NexHRIS is specifically developed to address the employee file management needs of the ISPSC HR Office.

The respondents of this study will consist of two groups. The first group is composed of the two (2) authorized users of the NexHRIS system within the HR Office, the HR Administrator and the HR Assistant, both of whom operate the system on behalf of the HR Office. Since the total number of HR Office users is only two (2), total enumeration will be applied for this group.

The second group consists of the teaching and non-teaching personnel of ISPSC Tagudin Campus with a total population of 142 employees. The researchers used Slovin's Formula to determine the appropriate sample size for this group. From a population size of 142, a sample size of 105 employees was chosen to represent the population with a 5% margin of error.

Slovin's Formula

$$n = N / [1 + N(e^2)]$$

Whereas: n = number of samples; N = total population; e = margin of error.

Given: N = 142; e = 0.05 (5%).

$$n = 142 / [1 + 142(0.05)^2] = 104.80 \approx \mathbf{105}$$

The 105 employee-respondents will be selected through simple random sampling, wherein each of the 142 employees will be given an equal chance of being selected until the required sample size is reached.

Table 2. Distribution of All Respondents

Respondents	Sample Size
HR Administrator	1
HR Assistant	1
Employees	105
TOTAL	107

Table 2 presents the distribution of all respondents. This study will involve a total of 107 respondents, composed of two (2) HR Office personnel, the HR Administrator and HR Assistant who will serve as the primary system operators, and 105 teaching and non-teaching personnel of ISPSC Tagudin Campus who will serve as employee-users of NexHRIS. The respondents were identified based on their direct involvement with or as intended users of the NexHRIS system, ensuring that both the administrative and employee perspectives are represented in the system evaluation.

Research Instruments

The data for this study will be collected through two methods. First, structured interviews will be conducted with the ISPSC HR Administrator prior to system development to determine the current employee file management processes, challenges, and requirements that will form the basis for the NexHRIS feature design. Second, after system deployment, the Technology Acceptance Model (TAM) questionnaire will be administered to all respondents — the HR Administrator, HR Assistant, and selected employee-users — to evaluate the acceptability of the system.

The Technology Acceptance Model (TAM), originally developed by Davis (1989), is a widely validated and widely used theoretical framework for evaluating user acceptance of information systems. TAM posits that two primary constructs, Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), determine a user's attitude toward using a system and, ultimately, their behavioral intention to use it. Perceived Usefulness refers to the degree to which a user believes that using the system will enhance their job performance, while Perceived Ease of Use refers to the degree to which a user

believes that the system will be free of effort. The TAM questionnaire to be used in this study will assess both constructs using a five-point Likert scale, with 1 indicating strong disagreement and 5 indicating strong agreement with each statement.

The TAM instrument is particularly appropriate for this study because NexHRIS is a new system being introduced to users who have no prior experience with a digital employee file management system. Evaluating the system through TAM will allow the researchers to determine whether the users find NexHRIS useful for their HR document management tasks and whether they find it easy to operate, which are the two most critical determinants of system adoption and continued use in institutional settings.

Data Analysis

The researchers will use appropriate statistical and qualitative tools to analyze the data gathered for each objective of the study. The following presents the data analysis methods to be applied per objective.

For Objective 1, which aims to analyze the current employee file management processes of the ISPSC HR Office, the qualitative data gathered through structured pre-development interviews with the HR Administrator and HR Assistant will be analyzed using thematic analysis. This involves systematically organizing and categorizing the responses into recurring themes related to the current HR document management workflow, the challenges and inefficiencies experienced by the HR Office, and the specific system features and requirements identified by the HR personnel. The themes derived from this analysis will directly serve as the basis for the system design, feature prioritization, and functional requirements of NexHRIS.

For Objective 2, which aims to determine the system requirements for the development of NexHRIS, the functional and non-functional requirements identified

during the requirements-planning interviews and consultations will be documented and organized into system specifications. These requirements will be validated with the HR Administrator and translated into the modules, user roles, and workflows of the system, ensuring that the developed system directly addresses the needs identified in Objective 1.

For Objective 3, which aims to evaluate the level of acceptability of NexHRIS based on the responses of the HR Administrator and the employees of ISPSC Tagudin Campus using the Technology Acceptance Model (TAM), the researchers will use mean and standard deviation to analyze the quantitative data gathered from the TAM questionnaire. These descriptive statistical tools will be used to determine the level of acceptability of NexHRIS as perceived by the HR Administrator, HR Assistant, and employee-users of ISPSC Tagudin Campus in terms of Perceived Usefulness (PU) and Perceived Ease of Use (PEOU).

Each item in the TAM questionnaire will be rated by the respondents using a five-point Likert scale as presented in Table 3.

Table 3. Likert Scale for the TAM Evaluation of NexHRIS

Scale	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

The computed mean scores for each TAM construct, particularly Perceived Usefulness and Perceived Ease of Use, will be interpreted using the mean range scale presented in Table 4. This interpretation will help determine the respondents' level of agreement with each construct and assess the overall acceptability of the proposed system.

Table 4. Mean Range Interpretation for TAM Evaluation of NexHRIS

Mean Range	Interpretation
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

For Objective 4, which aims to determine whether there is a significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use, the researchers will use the Wilcoxon Signed-Rank Test. This non-parametric statistical test is appropriate for comparing two related conditions — the manual process and the NexHRIS system — using the same group of respondents, particularly when the data is ordinal in nature, as in the case of Likert-scale responses. The respondents will be asked to rate both the current manual process and the NexHRIS system using the same TAM-based items on Perceived Usefulness and Perceived Ease of Use. The Wilcoxon Signed-Rank Test will then be applied to determine whether a statistically significant difference exists between the two sets of ratings. A significance level of 0.05 will be used as the decision threshold. If the computed p-value is less than 0.05, the null hypothesis that there is no significant difference between the manual process and NexHRIS will be rejected, indicating that a significant difference exists in favor of the developed system.

Chapter 3

RESULTS AND DISCUSSION

This chapter presents the findings of the study, focusing on the evaluation of the current employee management practices at the institution, the key features of the developed NexHRIS system, and the assessment of its acceptability. The results highlight how NexHRIS enhanced record security, streamlined HR transactions, and supported better accessibility and efficiency within the HR setting.

Findings

The following were the findings in relation to the interview that led to the conduct of the study:

The Current Process of Employee Records and HR Data Handling

The current HR process at ISPSC Tagudin Campus relied entirely on manual, paper-based and spreadsheet-based procedures. Employee 201 files, including the Personal Data Sheet, leave records, and other personnel documents, were kept and organized physically by the HR Office, with a Personal Data Sheet per employee, a separate form per leave application, and two handwritten ledger cards per person. Whenever an employee needed to update their Personal Data Sheet, they had to personally visit the HR Office, retrieve their physical file, and manually revise the necessary entries before having it re-filed. Leave applications followed a manual route in which a printed form was physically carried between three offices for signature, and employees were required to check their leave credits and file their requests in person, relying entirely on the HR personnel to locate their records, compute their balances, and process their applications by hand. This heavy reliance on physical filing made records vulnerable to loss, damage, and unauthorized access, while the absence of a

centralized digital repository required HR personnel to spend considerable time locating, reproducing, and verifying documents for every transaction. As the volume of personnel and administrative responsibilities at the institution continued to grow, this traditional method made the process time-consuming, prone to errors, and increasingly difficult to sustain and manage efficiently.

Features of the Developed System

NexHRIS moves the campus's HR transactions online while preserving the official Civil Service forms exactly as the campus uses them. The system is presented below by module, and where a module behaves differently for each role, the corresponding interfaces are shown separately.

Authentication and Role-Based Access

Users sign in with an email and password, with an optional two-factor code sent by email. After sign-in each account is redirected to the interface appropriate to its role. The four roles share one interface and differ only in what they may do, and their data boundaries are enforced in the query rather than the interface, so a Dean who edits a web address to reach another college receives a refusal rather than data.

[Plate 1. Login Page — insert screenshot/image here]

Plate 1. Login Page

[Plate 2. Two-Factor Verification — insert screenshot/image here]

Plate 2. Two-Factor Verification

Dashboards

Each role opens to a different set of figures. The HR Administrator sees campus-wide counts of pending reviews, submissions, and recent activity; the Campus Director sees campus-wide leave figures awaiting final signature; the Dean sees only the college

they hold, with the applications awaiting their first signature; and the Employee sees the status of their own leave, PDS, and ledger balances.

[Plate 3. HR Administrator Dashboard — insert screenshot/image here]

Plate 3. HR Administrator Dashboard

[Plate 4. Campus Director Dashboard — insert screenshot/image here]

Plate 4. Campus Director Dashboard

[Plate 5. Dean Dashboard — insert screenshot/image here]

Plate 5. Dean Dashboard

[Plate 6. Employee Dashboard — insert screenshot/image here]

Plate 6. Employee Dashboard

Employee Accounts, Colleges, and Departments

The HR Administrator creates and maintains staff accounts, assigning each employee a college and a department, where a department belongs to exactly one college. The college an employee is assigned to determines who signs that employee's leave. Employee number, position, college, department, and role remain with the HR Administrator, while every user edits their own name, email, contact number, photo, and password from My Profile.

[Plate 7. Employee Account Management — insert screenshot/image here]

Plate 7. Employee Account Management

[Plate 8. Colleges and Departments — insert screenshot/image here]

Plate 8. Colleges and Departments

[Plate 9. My Profile — insert screenshot/image here]

Plate 9. My Profile

Personal Data Sheet

The HR Administrator publishes the blank Civil Service form. An employee downloads it, fills it in, and uploads the completed workbook; the HR Administrator

then reads the submission and either approves it or returns it with remarks, and every version submitted is kept. Deans and the Campus Director submit their own Personal Data Sheet through the same employee interface. The exported document is a conversion of the workbook the employee actually uploaded rather than a form rebuilt from stored values, so the printed sheet is the campus's own form.

[Plate 10. Employee PDS Submission — insert screenshot/image here]

Plate 10. Employee PDS Submission

[Plate 11. HR Administrator PDS Review — insert screenshot/image here]

Plate 11. HR Administrator PDS Review

Leave Application and Approval Chain

An employee downloads the blank leave form, fills it in, and uploads it. The application is then routed according to the rank of the applicant, since no user signs their own leave. Each reviewer in turn reads the application as a PDF in the browser and either approves it or returns it with remarks, and a returned form restarts from the first stage. The Dean signs first for members of their own college, the HR Administrator second, and the Campus Director last; a Dean's own application skips the Dean stage, and the Campus Director's is signed by the HR Administrator alone. Once fully approved, the HR Administrator records the days against a ledger card and the employee may print the approval sheet.

[Plate 12. Employee Leave Filing — insert screenshot/image here]

Plate 12. Employee Leave Filing

[Plate 13. Employee Leave Status Tracking — insert screenshot/image here]

Plate 13. Employee Leave Status Tracking

[Plate 14. Dean Approval Queue — insert screenshot/image here]

Plate 14. Dean Approval Queue

[Plate 15. HR Administrator Approval Queue — insert screenshot/image here]

Plate 15. HR Administrator Approval Queue

[Plate 16. Campus Director Final Approval — insert screenshot/image here]

Plate 16. Campus Director Final Approval

[Plate 17. Printed Approval Trail with Skipped Stage — insert screenshot/image here]

Plate 17. Printed Approval Trail with Skipped Stage

Leave Calendar

The calendar shows who is on leave on a given date, scoped to the role viewing it. The Dean sees only their own college, while the HR Administrator and Campus Director see the whole campus.

[Plate 18. Leave Calendar (Dean, own college) — insert screenshot/image here]

Plate 18. Leave Calendar (Dean, own college)

[Plate 19. Leave Calendar (campus-wide) — insert screenshot/image here]

Plate 19. Leave Calendar (campus-wide)

Leave and Service Credit Ledger Cards

The campus keeps two cards per employee on the same official form. A day taken is charged either against the vacation or sick balance on the leave ledger, or against service credits on the service credit ledger, and the HR Administrator decides which card an approved leave is written on. A line on one card never appears on the other. Each line stores the balance as it stood after it, so correcting a line in the middle replays every balance below it. The HR Administrator maintains all cards; employees read their own.

[Plate 20. HR Administrator Ledger Maintenance — insert screenshot/image here]

Plate 20. HR Administrator Ledger Maintenance

[Plate 21. Employee Ledger View — insert screenshot/image here]

Plate 21. Employee Ledger View

HR Policies and Announcements

The HR Administrator publishes policies with acknowledgement tracking and posts announcements that reach users through a notification bell. Employees, Deans, and the Campus Director read published policies and acknowledge them.

[Plate 22. Policy Management and Acknowledgement Tracking — insert screenshot/image here]

Plate 22. Policy Management and Acknowledgement Tracking

[Plate 23. Employee Policy View — insert screenshot/image here]

Plate 23. Employee Policy View

Digital Identification Card

Each employee is issued a printable digital identification card bearing a QR code that resolves to a public verification page.

[Plate 24. Digital Identification Card — insert screenshot/image here]

Plate 24. Digital Identification Card

Audit Trail

Every sign-in, approval, and record change is recorded with the account and the IP address behind it, giving the HR Administrator a reliable trail for accountability.

[Plate 25. Audit Trail — insert screenshot/image here]

Plate 25. Audit Trail

To ensure data security, the system applies role-based access control that distinguishes the permissions of the HR Administrator, Campus Director, Dean, and Employee, and enforces these boundaries at the query level. With these functions, NexHRIS enhances the security, accessibility, and efficiency of employee records management at ISPSC Tagudin Campus.

Acceptability of the Developed System

This section presents the results of the Technology Acceptance Model (TAM) evaluation of NexHRIS in terms of Perceived Usefulness and Perceived Ease of Use, as well as the results of the Wilcoxon Signed-Rank Test comparing the existing manual process with the developed system. [The tables and figures in this section are to be completed after the TAM questionnaire has been administered and the data analyzed.]

[Table 5. Perceived Usefulness of NexHRIS (to be completed after data gathering) — insert screenshot/image here]

[Table 6. Perceived Ease of Use of NexHRIS (to be completed after data gathering) — insert screenshot/image here]

[Table 7. Wilcoxon Signed-Rank Test Results (to be completed after data gathering) — insert screenshot/image here]